Review plan · for the teacher

Models with numbers: unit check

Optimization · Level 2 · Unit check

Time	Materials
40 min	Board, chalk, paper. No devices needed.

Learning goals

- From words to a model
- The best plan is at a corner
- Whole numbers and yes-or-no choices
- Your first model in Python

Activity: Team quiz

1. Teams (3 min): Make teams of four and give each a sheet for its answers.
2. Quiz (15 min): Read each question aloud; teams talk it through and write one answer.
3. Explain (7 min): Reveal each answer and ask a different team to explain why it is right, using the words variable, constraint and objective.
4. Model it (10 min): Each team picks a school decision (duties, a purchase, the garden) and writes its variables, constraints and objective on paper.
5. Wrap-up (5 min): Write the question most teams missed on the board, and start tomorrow's lesson with it.

Discussion

- Which idea from this unit will you use first in class?
- Where could rounding mislead us at school?
- Who should help write the rules of a school model?

Exit ticket

- One decision I will model with my class, and its objective.

Try it at school tomorrow

Pick one decision at your school and write it as a model with a colleague: the variables with units, the constraints and the objective. Check one plan by putting in its numbers, and ask who the model leaves out.

Quiz sheet · for each team

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Team _____ Date _____

1 **In the school lunch model, which part is the objective?**

- ◯ Cups of rice per pupil
- ◯ The price of a cup of rice
- ◯ The total cost, to make as small as possible
- ◯ Iron at least 5 mg

2 **Match each piece of the lunch model to what it means.**

energy ≥ 700	• • • •	The cost, to minimise
nuts ≤ 2	• • • •	A variable: cups per pupil
cost = 15 × rice + … + 8 × nuts	• • • •	Enough energy
rice	• • • •	At most 2 handfuls of groundnuts

3 **The garden plan 6 tomato beds and 2 bean beds keeps all three limits (x + y ≤ 8, 2x + y ≤ 14, x + 3y ≤ 18).**

- ◯ True
- ◯ False

4 **Why does the simplex method only visit corners of the feasible region?**

- ◯ Plans inside the region break a limit
- ◯ Corners are easier to draw
- ◯ The best plan of a linear program is always at a corner

5 **Which decisions need whole-number or yes-or-no variables? Choose all that apply.**

- ☐ Whether to fund the chess club
- ☐ How many teachers to hire
- ☐ How many buses to hire
- ☐ Kilograms of rice for the kitchen
- ☐ Litres of paint for the classroom

6 **Complete the garden model: the land limit and the goal.**

```
model. _____(x + y _____ 8)
model.add(2 * x + y <= 14)
model.add(x + 3 * y <= 18)
model. _____(30 * x + 40 * y)
```

Word bank: `<=` `>=` `add` `solve` `maximize` `minimize`

7 **Put the parts of a CP-SAT program in order.**

- [ ] Solve with a solver
- [ ] Set the objective
- [ ] Make the model
- [ ] Read the values and check them
- [ ] Add the constraints
- [ ] Make the variables

8 **This program stops with an error before printing. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  model = cp_model.CpModel()
2  x = model.new_int_var(0, 10, "tomato")
3  model.add(2 * x <= 14)
4  model.maximize(30 * x)
5  solver = cp_model.CpSolver()
6  status = solver.solve()
7  print(solver.value(x))
```

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9 **What does this program print?**

```
x = 4
y = 4
print(x + 3 * y <= 18)
```

◯ False ◯ 16 ◯ True

10 **Complete the sentence about the trip buses.**

Rounding 2.13 buses _____ leaves four people behind; rounding _____ costs 1,650; the best _____-number plan costs 1,400.

Word bank: up · corner · fraction · down · whole

Answers · for the teacher

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1 **The total cost, to make as small as possible**

The objective is the one expression to minimise or maximise: here the cost.

2 **energy ≥ 700 — Enough energy**
nuts ≤ 2 — At most 2 handfuls of groundnuts
cost = 15 × rice + … + 8 × nuts — The cost, to minimise
rice — A variable: cups per pupil

A constraint has a sign (≥ for at least, ≤ for at most); the objective is an expression with no sign, to minimise; a variable is a single decision with a unit.

3 **True**

Land 6 + 2 = 8, water 12 + 2 = 14 and hours 6 + 6 = 12: every limit holds, so (6, 2) is feasible.

4 **The best plan of a linear program is always at a corner**

With a straight-line goal, from any point inside you can still improve until you reach an edge and then a corner.

5 **How many buses to hire; Whether to fund the chess club; How many teachers to hire**

Buses and teachers come whole, and funding a club is yes or no. Paint and rice can be measured in any amount, so fractions are fine there.

6

```
model.add(x + y <= 8)
model.add(2 * x + y <= 14)
model.add(x + 3 * y <= 18)
model.maximize(30 * x + 40 * y)
```

Each limit is model.add(…) with the right sign: at most 8 beds is <=.

7 **Top to bottom: 5, 4, 1, 6, 3, 2**

The variables belong to the model, and the constraints and objective are written with the variables.

8 **Line 6: `status = solver.solve(model)`**

The solver needs to be told which model to solve: solver.solve(model).

9 **True**

4 + 3 × 4 = 16, and 16 ≤ 18, so it prints True: the plan (4, 4) keeps the hours limit.

10 **Rounding 2.13 buses down leaves four people behind; rounding up costs 1,650; the best whole-number plan costs 1,400.**

Rounding down breaks the seating limit and rounding up costs more than needed.